

Cooler roofs · lower bills · clearer quotes

HACKATHON PROJECT

Cool Roofing Intelligence

Cost, comfort and energy-savings predictions before the first quote

What is cool roofing?

Heat-reflective roofing (coatings, sheets or tiles) that lowers roof-surface temperature, so buildings stay cooler and need less cooling electricity.

Objective and relevance

Give roofing companies data-backed answers before quoting: what will it cost, and what will the customer gain?

The solution

One tested web system that learns from 2,500 past installations to predict, compare and explain.

2,500
historical installations

5 tasks
6 trained models

8 pages
web app with PDF proposals

PROBLEM STATEMENT

Five questions a roofing customer asks, and the data to answer them

1

Installation cost

Total cost of the job in ₹

Regression

2

Roof temperature drop

Temperature before − after (°C)

Regression

3

Electricity savings

Monthly kWh before − after

Regression

4

Customer rating

1–5 stars; high = 4★ or more (our assumption)

Regression + classification

5

Customer referral

Did the customer refer the company? (yes/no)

Classification

Why historical installation data helps

- Each of the 2,500 past jobs records what was known at quote time and what happened afterwards.
- Models learn how size, roof, method and quality moved cost, cooling, savings and customer feedback.
- Result: consistent, explained estimates instead of guesswork, with limits stated.

Source: cool_roofing_problem.md; targets as defined in docs/data_dictionary.md

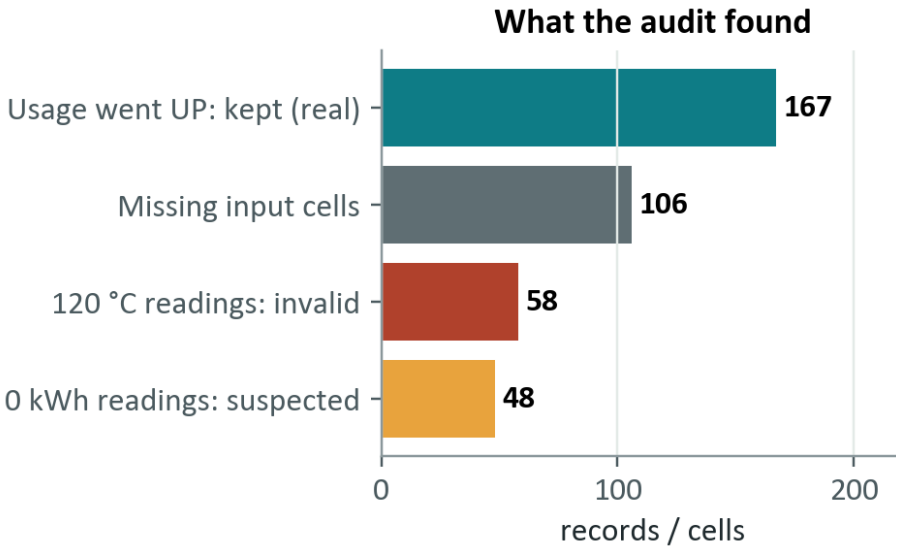
The audit found hidden placeholders; no data was deleted

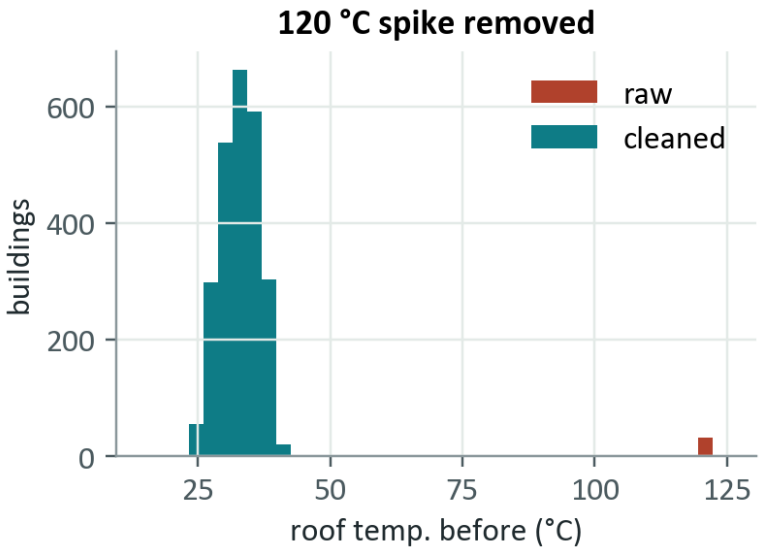
2,500 records · 15 columns

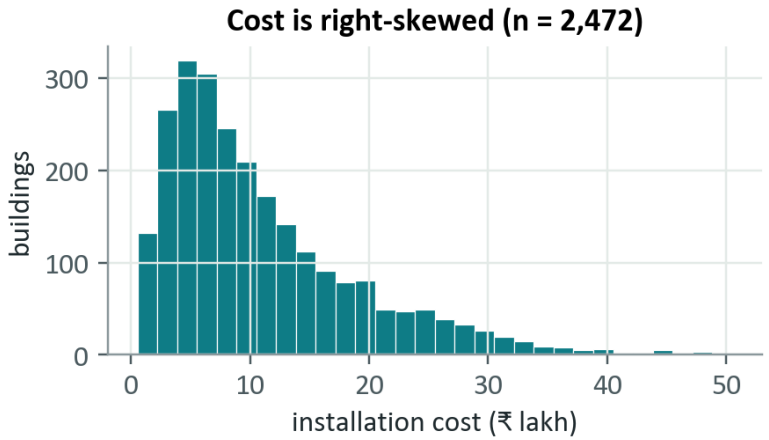
Inputs: location, size, age, existing roof, method, quality, start temp., AC units, current kWh

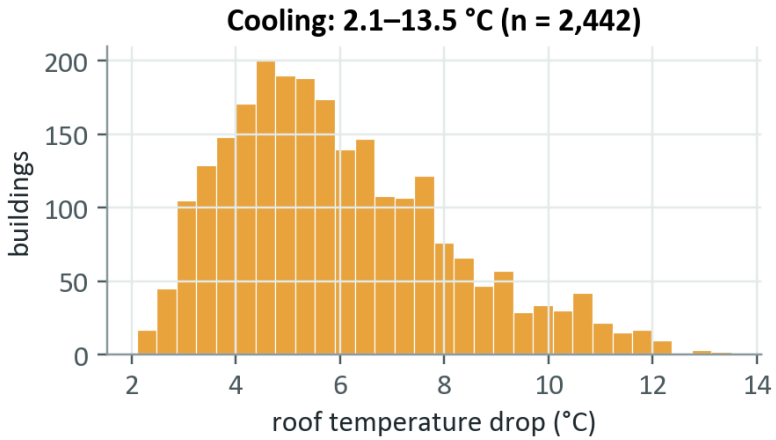
Targets: cost, temperature drop, kWh saved, rating, referral

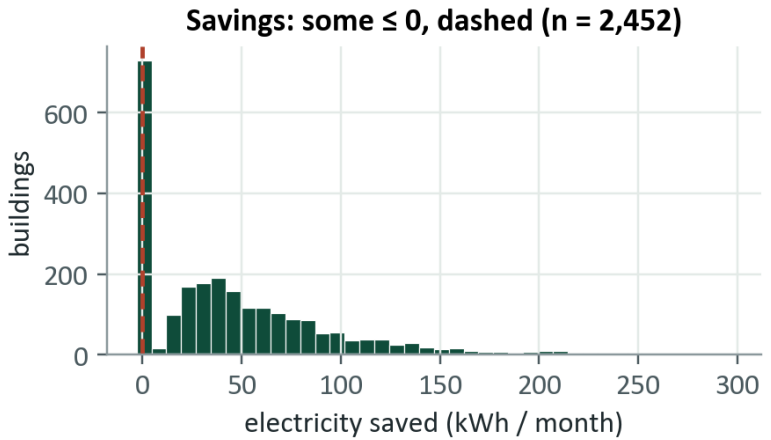
Decisions: 120 °C → missing; 0 kWh → missing (inferred); 167 increases kept; 106 gaps imputed in training only; 0 rows deleted











Effect of cleaning: electricity-savings hold-out MAE 4.45 → 3.62 kWh. Zero-kWh treatment is an inference; the 120 °C finding is confirmed.

Source: dataset/cool_roofing_data.csv (read-only); reports/data_quality_report.md, cleaning_audit.csv; charts from reports/data_profile.json

MODELS USED

Five tasks, six models, each chosen by validation evidence

Task	Output	Selected model	Why it was chosen
Installation cost	₹ (regression)	Log-linear Ridge	Cost ≈ size × rate: beat boosting in CV
Roof temperature drop	°C (regression)	LightGBM	Predicts a fraction of start temperature
Electricity savings	kWh/month (regression)	LightGBM	Predicts a fraction of current usage
Customer rating	1–5 stars (regression)	LightGBM	Best CV error; clipped to 1–5
High rating (≥ 4★)	yes / no (classification)	LightGBM	Best CV accuracy
Customer referral	yes / no (classification)	Logistic Regression	Best CV F1; weak signal overall

Regression: predicts a number

Cost (₹), temperature drop (°C), electricity saved (kWh), rating (stars).

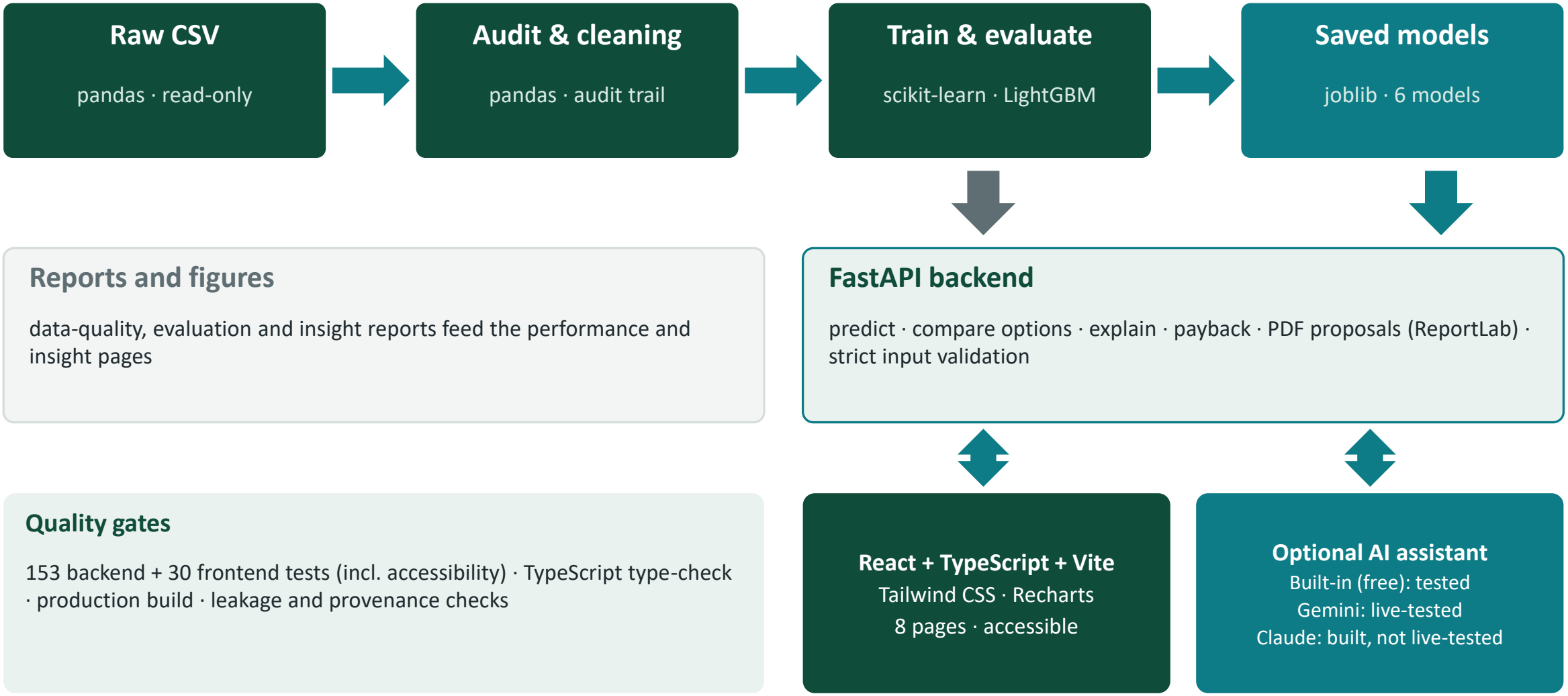
Classification: predicts a yes / no

High rating (≥ 4★) and customer referral, with a probability.

Source: models/model_metadata.json (selected by cross-validated score; baselines are never selectable)

TECHNOLOGY STACK

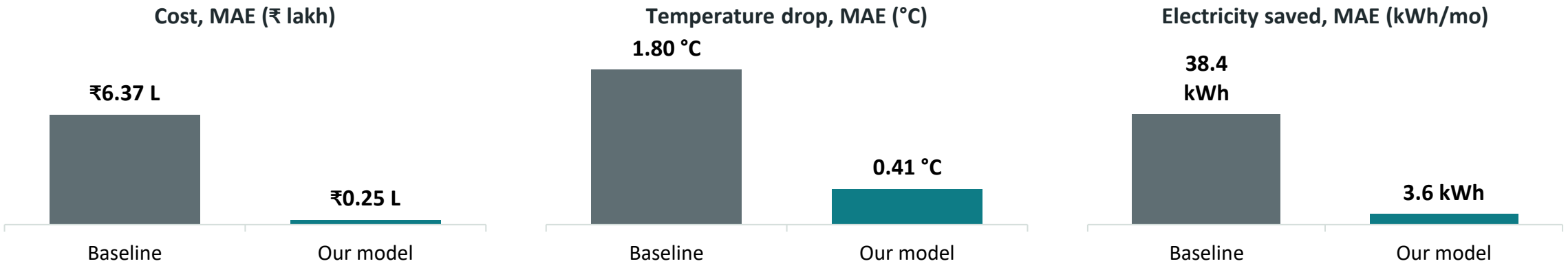
A tested web product built on a standard, open technology stack



Source: requirements.txt, package.json, docs/architecture.md; 153 backend + 30 UI tests last run

Strong on cost, cooling and savings; weak on rating and referral

MAE average miss · **RMSE** big misses weigh more · **R^2** variation explained · **Accuracy** % correct · **F1** precision/recall balance · **ROC-AUC** ranking skill (0.5 = chance)



Task	MAE	RMSE	R^2	Baseline MAE
Installation cost	₹25,299	₹38,515	0.998	₹637,024
Roof temperature drop	0.41 °C	0.51 °C	0.948	1.80 °C
Electricity savings	3.62 kWh	5.87 kWh	0.987	38.4 kWh

Task	Our model	Trivial baseline	Verdict
Rating (MAE)	0.77 ★ · R^2 0.36	0.92 ★ (median)	better
High rating ≥4★	acc 68.8% · F1 0.614 · AUC 0.76	acc 57.6% · F1 0.596	acc better · F1 n.s.
Referral	F1 0.569 · acc 51.2% · AUC 0.66	F1 0.544 · acc 62.6%	F1 n.s. · acc worse

Held-out evaluation

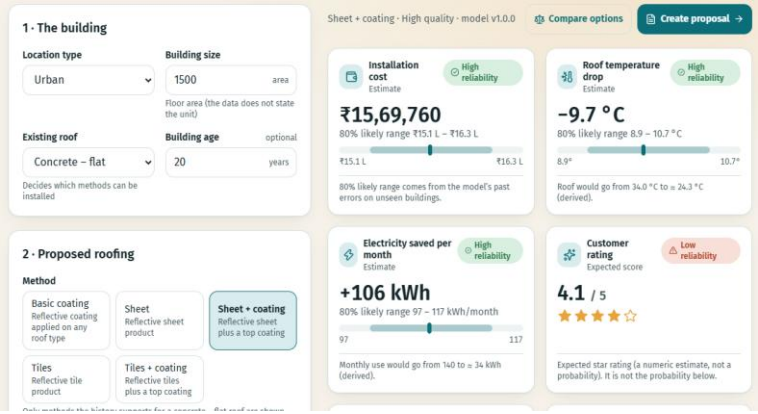
500 buildings set aside before modelling, scored once. Baselines scored on the same rows. 'Better' only if the 95% bootstrap interval excludes 0; n.s. = not statistically distinguishable.

Why rating/referral are weaker

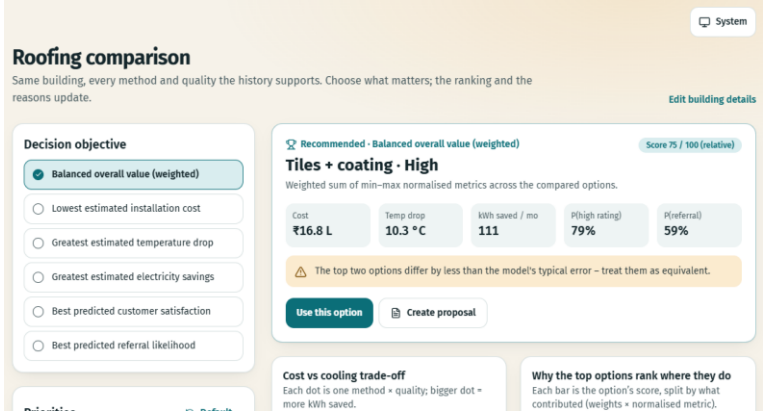
They depend on experience after installation, not known at quote time. Use to rank, not promise.

Source: models/model_metadata.json, reports/significance.json. All results on 500 held-out buildings never used in training or tuning.

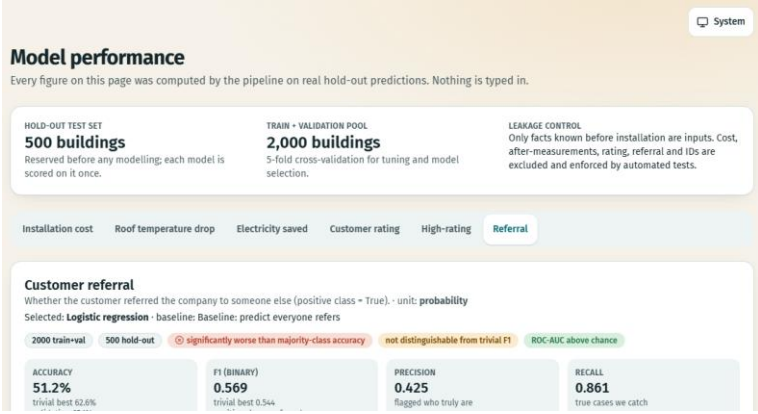
Working software: predict, compare, check quality, propose



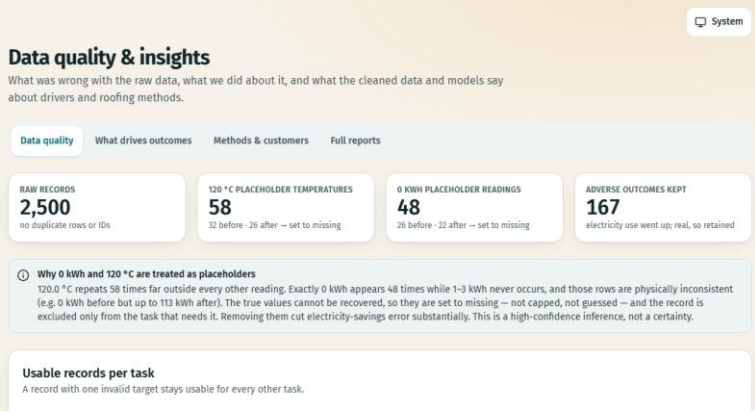
Customer prediction: cost, cooling, kWh, rating, referral



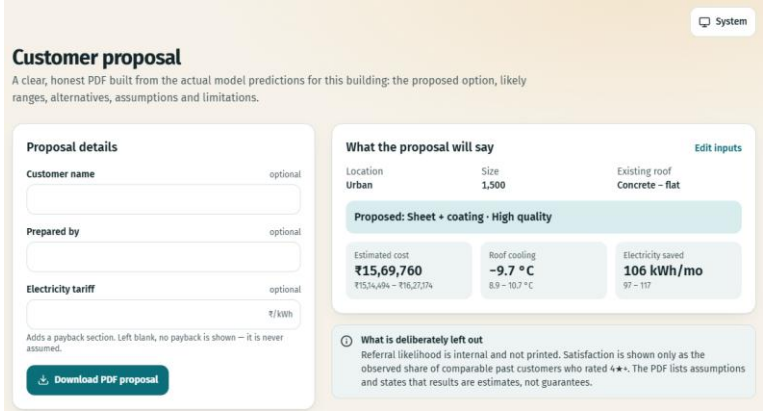
Option comparison: 15 method × quality options ranked



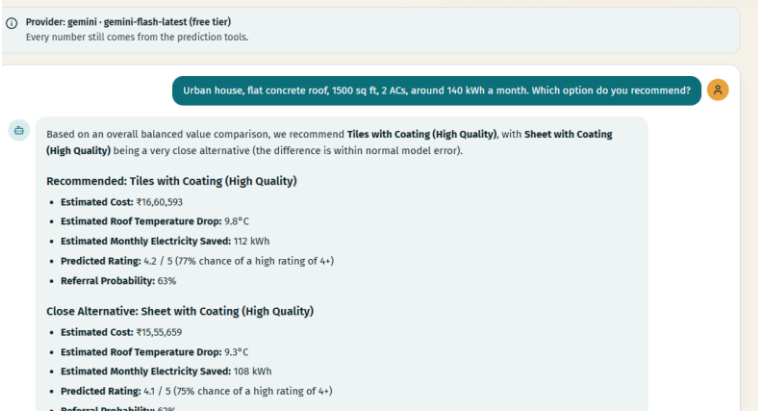
Model performance: baselines, intervals, confusion matrix



Data-quality insights: audit counts and cleaning decisions



PDF proposal: built from real predictions



AI assistant: Gemini live-tested; Claude untested

Business value: prepare preliminary estimates in seconds, compare options before quoting, hand over a proposal with its assumptions stated.

Source: screenshots of the running local application. Assistant: Gemini live-tested; Claude built but not live-tested.

Explained, honest estimates from historical roofing data

What the project achieves

- Audited 2,500-record dataset; 0 rows deleted
- 5 tasks, 6 models, held-out evaluation
- Cost MAE ₹25,299
- Cooling MAE 0.41 °C · savings MAE 3.62 kWh
- Compare options · explain · PDF proposals

Value and limitations

- Faster preliminary estimates; options compared before quoting
- Dataset: no time or location fields; 0 kWh fix is inferred
- Rating and referral only weakly predictable
- High rating = 4★ is our assumption
- Estimates, not guarantees

Future scope (not implemented)

- Validate on newly measured installations
- Digital-twin simulator
- Carbon-savings tracker
- Geo-climate intelligence

Historical roofing data, turned into estimates a sales rep can trust, with the reasons and the limits attached.

Thank you · Questions?